

Multimodal Flow Matching for Probabilistic Geolocalization

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CS-503 Project Proposal

Abstract—Image geolocalization aims to predict where an image was captured. Most existing methods are deterministic and do not explicitly model spatial ambiguity. Recent generative approaches address this limitation by modeling distributions over possible locations through sampling, but rely only on visual conditioning. We adopt a flow matching approach, where random points on the Earth’s surface are guided toward plausible locations conditioned on the input image. We extend this framework with caption conditioning to introduce explicit semantic information, and evaluate whether additional modalities improve localization accuracy and reduce ambiguity in the predicted spatial distributions.

I. INTRODUCTION

Image geolocation is important for various tasks such as emergency responses, navigation, and urban analysis [1]. However, in reality, geolocation metadata is frequently missing for reasons like security. This data gap can be filled by image geolocalization—a method of determining the location given an image [2]. In this task, street-view images are particularly valuable by offering rich clues from a ground-level human-centered perspective.

Previously, image geolocalization often relied on matching a query image against geo-tagged databases or assigning it to predefined spatial partitions, facing hurdles like sparse reference coverage and coarse geo-partitions. Furthermore, these methods are deterministic and often rely on a single modality of information. Recent research has shifted toward generative multimodal learning, which can model spatial ambiguity and integrate auxiliary signals as well. This motivates a systematic study of how different data modalities and modeling choices can be combined to make image geolocalization more accurate and robust.

A. What is the problem you want to solve?

- 1) **Modality Ablation:** How do different modalities contribute to image geolocalization? Specifically, we evaluate whether textual descriptions and potentially aerial imagery improve performance beyond street-view imagery alone, and which signals are most informative.
- 2) **Methods Comparison:** How do deterministic, generative, and hybrid approaches perform in image geolocalization? In particular, we compare flow matching-based generative models with classification and hybrid baselines to understand their strengths and limitations.
- 3) **Generative Pipeline Design:** How do design choices in a generative geolocalization pipeline affect performance? In particular, how do different conditioning

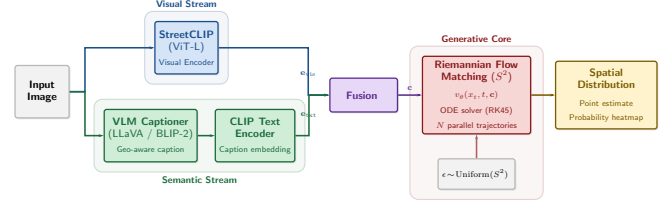


Figure 1. Multimodal probabilistic image geolocalization pipeline.

strategies and sampling settings influence the quality and sharpness of predicted spatial distributions, and how can these predictions be effectively visualized?

B. Why is it important that this problem be solved?

- 1) Through ablation study, we can understand how different modalities contribute in geolocalization, and whether models rely more on visual features or scene and semantic texts. It can also reveal if pre-trained embeddings capture robust and transferable geo-representations and what they have already learned.
- 2) Better image geolocalization methods can support large-scale image geotagging and help expand datasets’ spatial coverage, especially in underrepresented regions. It can further support the development of globally representative geo-foundation models.
- 3) Comparing different methods and pipelines can help us better understand deterministic and generative alternatives, providing practical guidance for building image geolocalization pipelines.

II. METHOD AND DELIVERIES

A. Model and Pipeline Architecture

The proposed system is a multimodal, probabilistic geolocalization pipeline that generates samples on the Earth’s surface, which are aggregated into spatial heatmaps (Figure 1).

The pipeline operates on two parallel input streams. A **StreetCLIP** [3] (ViT-L) backbone encodes the input image into a d -dimensional visual embedding. Concurrently, a VLM (**LLaVA** [4] or **BLIP-2** [5]) generates a structured caption explicitly describing geographic cues (e.g. climate, biome, traffic orientation, visible language), which is then encoded by a **CLIP Text Encoder** [6]. The two embeddings are merged to produce a unified conditioning vector $\mathbf{c}$.

The Earth is modelled as a unit sphere $\mathcal{M} = S^2$. A neural network modeling $v_\theta(x_t, t, \mathbf{c})$ conditioned on $\mathbf{c}$, learns a vector field over S^2 that transports random noise coordinates

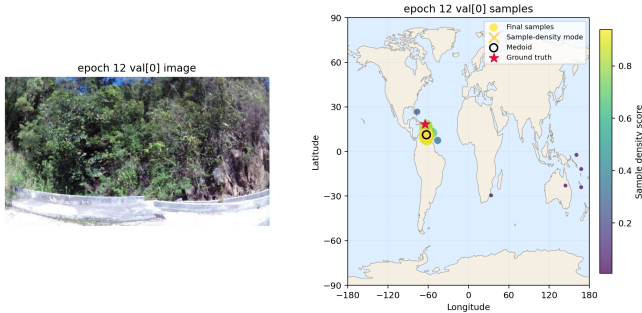


Figure 2. Prediction example with a proof of concept model

toward the true geographic location. At inference, N independent trajectories are integrated in parallel via an ODE solver, and their endpoints collectively form the predicted spatial probability distribution.

B. Evaluation Metrics

- **Deterministic Point-Estimate Metrics:** Haversine Distance: Median distance error between the predicted coordinate and the ground truth location; Top-K Distance Accuracy: the absolute percentage of predictions that fall within specific, cascading geographic radii (e.g. 200km) from the true location. GeoScore: A score inspired by the game GeoGuessr, which ranges from 0 to 5000, with higher scores indicating better accuracy.
- **Probabilistic and Density Estimation Metrics:** Negative Log-Likelihood (NLL): how much probability mass the model assigns to the true location; Spatial Entropy (Localizability): the Shannon entropy of the generated spatial distribution, the analysis can definitively quantify the “localizability” of an image.

C. Experiments and Timeline

- 1) **Multimodal Modeling Prototype Quick Run** (April 5th - 12th): A quick run of the multimodal flow matching prototype on a subset of the OSV-5M [7] dataset to show the viability of the model architecture and training targets (Figure 2).
- 2) **Multimodal Model Training with Verification on Inference** (April 13th - 28th): Train the complete model (image encoder + text encoder $\rightarrow$ Cross-Attention Fusion $\rightarrow$ RFM network). The goal is to scale the training pipeline correctly on the large dataset and check the both quantitative predicted accuracy and qualitative predicted distribution on the test data. The success metric requires the Haversine Distance to be less than 2800km, and NLL to be smaller than 0.0.
- 3) **Modality Ablation Study** (April 29th - May 20th): We will train three distinct variants for 50 epochs: Image-Only conditioning, Caption-Only conditioning, and Image+Caption conditioning. The primary success metric is achieved if the Image+Caption model demonstrates a statistically significant reduction in NLL and Top-100km distance error compared to the unimodal baselines on the 2k validation set.

The immediate, pre-planned fallback is to abandon the continuous ODE solver and revert entirely to the discriminative, hybrid architecture outlined in the PIGEON model [8].

III. RELATED WORK

Retrieval-based methods match images against large geo-tagged databases using cross-view or correlation-aware representations [9], [10], while classification approaches discretize the Earth into regions and predict a class label [11]. Hybrid models combine both [8]. Despite strong performance, these methods produce deterministic predictions and fail to capture ambiguity, as visually similar scenes can correspond to distant geographic locations.

Semantic signals have been introduced to improve generalization. Models such as StreetCLIP and PIGEON leverage synthetic captions or metadata [3], [8], but remain discriminative and output point estimates.

Generative approaches model geolocation as a spatial distribution. Dufour et al. [12] use diffusion and flow matching, while LocDiff [13] applies latent diffusion. These methods capture ambiguity but rely solely on visual conditioning. The role of text in this setting remains unclear: how to structure it for discriminative cues, whether it adds information beyond visual encoders, and whether gains reflect signal or redundancy. We address this by integrating text into a flow-matching model and systematically comparing text representations and their impact on accuracy and predicted distributions.

IV. DISCUSSION

A. Implications and Contribution

Our primary contribution is a generative geolocalization framework that utilizes VLM-generated geo-aware captions as explicit conditioning for flow-matching networks. We validate this approach through a multimodal ablation study comparing image, caption, and hybrid modalities. Broadly, this work provides empirical evidence for embedding semantic understanding into future geo-foundation models. Furthermore, we introduce spatial entropy as a practical confidence metric for high-stakes deployments.

B. Feasibility & Risk Assessment

This project is highly feasible, utilizing the publicly accessible OpenStreetView-5M dataset.

Preliminary tests show the base architecture runs successfully on a single V100 GPU, leading us to estimate the full multimodal pipeline can be trained using four V100 GPUs, with wall-clock time depending on the final dataset scale.

We foresee two primary risks. First, global data imbalances could induce prior biases, over-predicting densely sampled regions while yielding coarse representations for underrepresented areas. We plan to mitigate this through geographically stratified sampling or by initially constraining

the prediction space to data-rich regions like Europe. Second, VLM captions act as a lossy compression that might discard fine-grained visual details, potentially introducing noise rather than complementary signals. We will empirically evaluate the true informational utility of these textual descriptions through our proposed modality ablation studies.

REFERENCES

- [1] M. J. Bianco, D. Eigen, and M. Gormish, “Enhancing worldwide image geolocation by ensembling satellite-based ground-level attribute predictors,” in *2025 IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*, 2025, pp. 535–543.
- [2] Z. Zhou, J. Zhang, Z. Guan, M. Hu, N. Lao, L. Mu, S. Li, and G. Mai, “Img2loc: Revisiting image geolocalization using multi-modality foundation models and image-based retrieval-augmented generation,” in *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, ser. SIGIR ’24. New York, NY, USA: Association for Computing Machinery, 2024, p. 2749–2754. [Online]. Available: <https://doi.org/10.1145/3626772.3657673>
- [3] L. Haas, S. Alberti, and M. Skreta, “Learning generalized zero-shot learners for open-domain image geolocalization,” 2023. [Online]. Available: <https://arxiv.org/abs/2302.00275>
- [4] H. Liu, C. Li, Q. Wu, and Y. J. Lee, “Visual instruction tuning,” 2023. [Online]. Available: <https://arxiv.org/abs/2304.08485>
- [5] J. Li, D. Li, S. Savarese, and S. Hoi, “Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models,” 2023. [Online]. Available: <https://arxiv.org/abs/2301.12597>
- [6] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, “Learning transferable visual models from natural language supervision,” 2021. [Online]. Available: <https://arxiv.org/abs/2103.00020>
- [7] G. Astruc, N. Dufour, I. Siglidis, C. Aronsson, N. Bouia, S. Fu, R. Loiseau, V. N. Nguyen, C. Raude, E. Vincent, L. XU, H. Zhou, and L. Landrieu, “Openstreetview-5m: The many roads to global visual geolocation,” 2024. [Online]. Available: <https://arxiv.org/abs/2404.18873>
- [8] L. Haas, M. Skreta, S. Alberti, and C. Finn, “Pigeon: Predicting image geolocations,” 2024. [Online]. Available: <https://arxiv.org/abs/2307.05845>
- [9] S. Hu, M. Feng, R. M. H. Nguyen, and G. H. Lee, “Cvm-net: Cross-view matching network for image-based ground-to-aerial geo-localization,” in *The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2018.
- [10] F. Lu, X. Lan, L. Zhang, D. Jiang, Y. Wang, and C. Yuan, “Cricavpr: Cross-image correlation-aware representation learning for visual place recognition,” 2024. [Online]. Available: <https://arxiv.org/abs/2402.19231>
- [11] B. Clark, A. Kerrigan, P. P. Kulkarni, V. V. Cepeda, and M. Shah, “Where we are and what we’re looking at: Query based worldwide image geo-localization using hierarchies and scenes,” 2023. [Online]. Available: <https://arxiv.org/abs/2303.04249>
- [12] N. Dufour, V. Kalogeiton, D. Picard, and L. Landrieu, “Around the world in 80 timesteps: A generative approach to global visual geolocation,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2025, pp. 23 016–23 026.
- [13] Z. Wang, Z. Liu, J. Zhang, Z. Zhou, Q. Cao, N. Wu, L. Mu, Y. Song, Y. Xie, N. Lao, and G. Mai, “Locdiff: Identifying locations on earth by diffusing in the hilbert space,” 2025. [Online]. Available: <https://arxiv.org/abs/2503.18142>